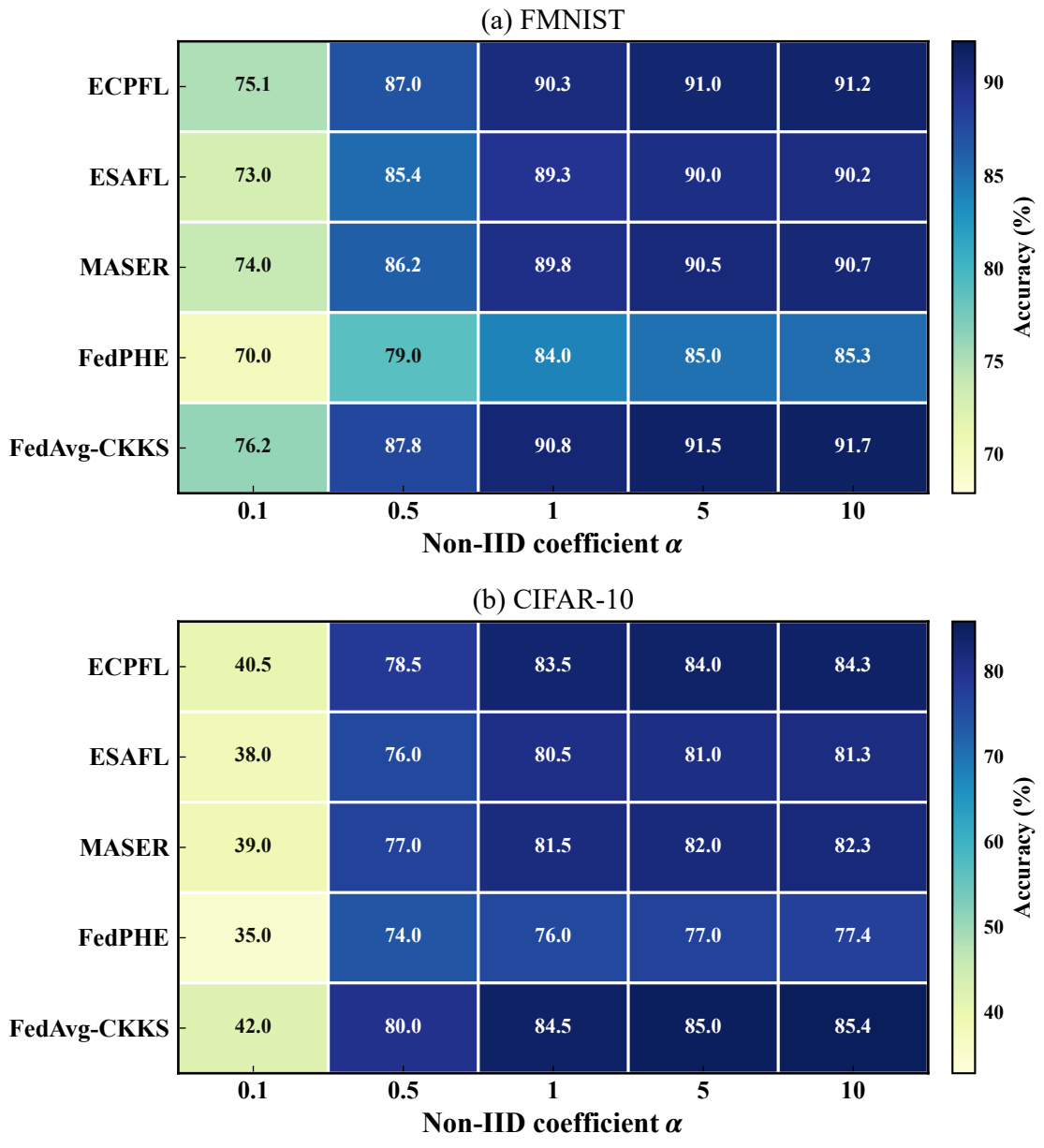

(a) FMNIST
ECPFL
ESAFL
MASER
FedPHE
FedAvg-CKKS
75.1
87.0
90.3
91.0
91.2
73.0
85.4
89.3
90.0
90.2
74.0
86.2
89.8
90.5
90.7
70.0
79.0
84.0
85.0
85.3
76.2
87.8
90.8
91.5
91.7
0.1
0.5
1
5
10
Non-IID coefficient α
Accuracy (%)
90
85
80
75
70
(b) CIFAR-10
ECPFL
ESAFL
MASER
FedPHE
FedAvg-CKKS
40.5
78.5
83.5
84.0
84.3
38.0
76.0
80.5
81.0
81.3
39.0
77.0
81.5
82.0
82.3
35.0
74.0
76.0
77.0
77.4
42.0
80.0
84.5
85.0
85.4
0.1
0.5
1
5
10
Non-IID coefficient α
Accuracy (%)
80
70
60
50
40